

14/1/2024

Biology

1. (3) tertiary consumer
2. (4) The concept is that hereditary information comes in discrete units.
3. (3) ~~de~~ decreasing the rate of ozone depletion
4. (i) primary consumers are mice and rabbit
(ii) the population of mice will increase and ~~seeds~~ population of seeds will decrease.
5. Mendel selected pea plant for experiments due to following reasons
 - it has 7 allelic characteristics with only 2 alleles for each characteristic
 - lifetime is short, therefore the result is given very early.
 - the flowers are bisexual, therefore there are both the options of pollination (cross or self).
6. (i) female (XX)
(ii) Variation help to promote survival of species. ~~by~~ Species live in a particular niche, niche can change anytime and it not in the species' control, therefore if the niche changes then the species would die unless there is a particular variation that is present in some of the ^{organism} ~~species~~ to live in the changed niche. Therefore the organisms having the variations would survive and prevent ~~extinction~~ dying out of the species.

(iii) Let the allele for pink be P and for white be p .
 ∴ If homozygous dominant is cross with homozygous recessive then,

P_1 gen gametes .

PP	$\times$	pp
(P) (P)		p p

	P	P	Phenotype = <u>all pink flowers</u>
p	Pp	Pp	
p	Pp	Pp	

If F_1 generation is cross,

F_1 gen gametes

Pp	$\times$	Pp
(P) (p)		(P) (p)

	P	p	Phenotypic ratio = Pink : White = <u><u>3 : 1</u></u>
P	PP	Pp	
p	Pp	pp	